Bayer Business Services

Service Delivery Contract
Mobile App Implementation
<App Title>

Version X.X as of DD.MM.YYYY

to Master Service Agreement (MSA) as of DD.MM.YYYY

Agreement between

Bayer Business Services GmbH
Hauptverwaltung
Gebäude B151
51368 Leverkusen
(hereinafter referred to as “Client” or „BBS”)

and

X
X
X
(hereinafter referred to as “Supplier”)

(jointly and separately hereinafter referred to as “Contract Partners”)

Number of pages (including this): **32**

This page must be deleted before sending the document to the Supplier.

DOCUMENT GUIDELINE

The purpose of this document is to support the complete demand to contract process.

To make best use of this template the BBS Service Owner is recommended to use section 2 and the guidelines therein to describe all functional and non-functional requirements.

The BBS Service Owner can then select appropriate statements in section 4 (Statement of Work) out of the statements marked with “optional”. After the selection the respective columns should be deleted.

Supplier competence can be utilized to either jointly discuss and align on the requirements or request respective clarifications or revised versions as part of a RfP response.

In any case it is recommended to align and explicitly agree on BBS requirements set out in section 2 and Supplier specifications set out in section 3 at least with two or with the selected Suppliers before project assignment.

REVISION HISTORY

Date	Version	Description	Contributor
DD.MM.YYYY	0.1	Draft	
DD.MM.YYYY	1.0	Final contract	
DD.MM.YYYY	1.X	First minor contract changes	
DD.MM.YYYY	2.0	Larger/major contract changes	

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1.0 INTRODUCTION

1. This Statement of Work (“SOW” or “App Design and Build Agreement”) is concluded under the above specified Master Service Agreement (“MSA”).

   Alternative: In case of no MSA

   This Statement of Work (“SOW”) is based on the respective “Conditions of Purchase“ and the „General Terms for Bayer AG and its holding companies for the purchase of IT consulting services including IT training for the client’s employees“ of the Client.

2. This SOW shall specifiy the obligations of Supplier resulting from the subject matter of this SOW, the agreed Service levels and the Compensation. The Contract Partners agree that Supplier shall perform the Services in accordance with the service description specified in the SOW.

2.0 FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

<Note to the BBS Service Owner: Create a clear and well defined description of the Mobile App Implementation requirements and their business and technical context. The requirements definition could be structured in two parts. Consider if the Supplier is required to support this requirements definition and in this case which of the following parts or aspects you would require the Supplier to describe.>

2.1 Main Functional Requirements

Description of context and main functionality considering but not necessarily limited to the following aspects:

1. Motivation and Mobile App objectives
2. Type of App (e.g., Medical / non-medical; B2B, B2C, B2E) and respectively intended user groups
3. Main functionality from a business perspective
4. Examples and pictures of key functionality if applicable
5. Particular social media requirements; required media and content elements
6. …. Any other topics that are relevant

2.2 Non-functional Requirements

Description of non-functionality considering but not necessarily limited to the following aspects:

1. List of smart phones and tablet models and platform versions to be supported (iPhone, iPad, iOS, Android, Windows etc., herewithin referred to as In-Scope-Devices)
2. Architectural, scalability and performance requirements
3. Any technology requirements resp. restrictions
4. Key design concept
5. Usability requirements and user input needs

6. Interfaces to enterprise systems and integration to back-end services
7. Reporting and mobile analytics requirements
8. Data synchronization and replication, data input and output requirements
9. Security and data privacy requirements

2.3 Detailed Functional Requirements (Use Cases)

<Note to the BBS Service Owner: To start with the Use Case description it may be useful to start with the description of the main functionality in part 1 of this section and discuss for each particular possbile result the previous steps the user had to perform to achieve this result. In a second step functions and steps which can be considered to provide self-contained usefull information or results can grouped accordingly. Each particular result should be described in a Use Case. A group of Use Cases reflecting a group of functions and steps as described above can be considered to reflect a mobile app screen. The Supplier is at least required to review - if deemed necessary to revise - the Use Cases and to obtain the BBS Service Owner's approval.>
Detailed description of User Case, i.e. each particular function provided on the intended devices' displays (see Annexure 4 containing the Use Case template and Annexure 5 as placeholder for the Use Cases).

3.0 FUNCTIONAL AND NON-FUNCTIONAL SPECIFICATION

3.1 High level architecture and technical feasibility study (proof of concept)

<Note to the Supplier: At minimum for each of the aspects listed below specify the respective app key design components. If the requirements as set out above are missing aspects required to describe a complete high level architecture or decide on the technical feasbility advice the Client accordingly and make respective recommendations to complete the requirements accordingly. >

1. Description of functional and non-functional specification considering the following aspects:

   a. In-Scope-Devices

   b. High level design to meet scalability and performance requirements

   c. Additional technology specifications considering Client restrictions

   d. High Level Data Model

   e. Description of usability standards to be used and specifications to meet the requirements for usability and user input needs considering the intended user groups

   f. Access technique and format to required media and specification of content usage

   g. Access to required data sources other than enterprise system and back-end services, respective interfaces and required format of data input and output / expected results

   h. Data synchronization and replication, data input and output techniques, methods and formats

   i. Interfaces to enterprise systems and integration to back-end services

j. Security and data privacy techniques and methods

k. Reporting and mobile analytics techniques and methods

l. Other aspects (please specify)

2. Comprehensive description of required tools, frameworks or standards:

   a. The usage of the SAP Mobile Platform (SMP) is deemed mandatory. In case Supplier adduces compelling reasons or Client has explicitly asked the Supplier to use a different mobile development platform the Supplier will specify the used platform functionality e.g. for backend access, user authorization etc. herewithin and obtain Client's written approval for the usage.

   b. Coding Standards & guidelines (if not covered by Webguideline)

   c. Planned mobility lab setup

   d. Devices and tools for development and testing (incl. Client Corporate Devices)

   e. Application Component Management Tool

   f. Code Assurance Checklists

   g. Code Generation toolkits & IDE's

   h. Planned reusable components, in particular Third Party or open source libraries

   i. Incident repository

   j. FAQ's & Checklists for development and testing

   k. Other …

3.2 User Interface / Screen specifications

<Note to the Supplier: The Supplier is required to review Uses Cases provided by Client with regard to completeness, consistency and clarity in order to cover the main functionality described in section 2. The Supplier shall advice the Client respectively and provide revised Uses Cases and additonal recommendations as part the RfP response. >

Description of all user interfaces / screens with reference to the Use Cases (each Use Case is to be referred to at least once).

4.0 STATEMENT OF WORK

4.1 Preamble

1. The Supplier´s responsibility includes the design and delivery of the App described in the previous section in a structured manner as a project covering the lifecycle of the App. In particular Supplier will deliver Software Development Lifecycle (SDLC) Services for each phase as detailed below. This document covers the phases of Design, Develop, Test and Publish as shown in the Figure below.

Figure 1: Mobile App Lifecycle Phases

2. Furthermore the Supplier´s shall adhere to the standards set out in the Bayer Group Webguide (see Attachment; hereinafter refrerred to as Webguide) which builds an integral part of this contract and the basis of this Statement of Work. The Webguide is a living document and will be updated from time to time to reflect technology changes and standards across Bayer. The Supplier shall ensure to always work according to the actual version which is published in the Bayer intranet.

4.2 Implement - Design

Category	Services	Classification (Mandatory / Optional)
Topic	Overall Architecture	M
Scope	• Final architecture description covering scalability, performance and technology specification	M
	• Comply with state-of-the-art App development methods, data security standards and in addition to the Webguide other programming or design standards approved by the Client	M
	• Obtain agreement from the Client for all standards and methods other than set out in the Webguide which are planned for usage.	M
Deliverables	Overall Architecture documentation (The Overall Architecture documentation builds the frame for the final Architecture and Design documentation which consists of the detailed design documentation as set out in this section)	
Acceptance Criteria / KPI (if applicable)	• Approved technical feasibility Confirmed scope by the Client	

Category	Services	Classification (Mandatory / Optional)
Topic	Detailed Design Hardware platform and architecture	M

Category	Services	Classification (Mandatory / Optional)
Scope	• Detailed design specifying all required adaptations and modification required to provide the functionality and the specified themes on the In-Scope-Devices considering respective device-specific user experience	M
	• The App architecture shall conform to the latest technologies and be compatible with the market relevant App stores for easy distribution.	M
Deliverables	• Design document • Architecture design	
Acceptance Criteria / KPI (if applicable)	• App should be designed to run on all major devices released in last 18 months	

Category	Services	Classification (Mandatory / Optional)
Topic	Detailed content and user design	M
Scope	• Detailed data model specifying the data and content usage from the Client internal and external sources in scope (apart from Enterprise systems and back-end services).	M
	• Detailed design specifying the user interfaces with the same "look and feel" for the In-Scope-Devices providing a self-explanatory and intuitive handling for the users with regard to the intended user groups	M
	• Optimize the functionality design to optimally use the In-Scope-Devices' screens and resolutions and the keypad types (e.g. virtual, hard, touch pad)	O
	• The user interface design shall encourage and facilitate habitual usage e.g. providing measures such as gamification	O
Deliverables	• User interface design / wireframe • Wireframe handling walkthroughs (verifying ease of use, self-explanatory and intuitive handling)	
Acceptance Criteria / KPI (if applicable)	• # of clicks / user interactions required for user to navigate from App opening screen down to last functionality • Wireframe Walkthrough approval	

Category	Services	Classification (Mandatory / Optional)

Category	Services	Classification (Mandatory / Optional)
Topic	App icon	M
Scope	• Design an application icon that meets the standards and design guidelines of the Client.	M
	• Design an application icon that meets the recommended icon guidelines for the different App stores.	O
Deliverables	• App Icon	
Acceptance Criteria / KPI (if applicable)	• Approval from the Client • Acceptance from App stores	

Category	Services	Classification (Mandatory / Optional)
Topic	Mobile Analytics	O

Scope	• Detailed design specifying the monitoring of the following parameters during the usage of the mobile app with the least amount of implementation and evaluation effort:
	  • Active users
	  • Visit frequency
	  • Session time
	  • Depth of visit
	  • Revenue per user
	  • Social shares
	  • Retention rate
	  • Acquisition cost
	  • User transactions	

Deliverables	• User interface design / wireframe	
Acceptance Criteria / KPI (if applicable)	• Wireframe Walkthrough approval	

Category	Services	Classification (Mandatory / Optional)
Topic	Data synchronization and replication	O
Scope	• The App shall work regardless of the user's connectivity to the internet, as well as give users the option to download and save content to a personal "reading list" for access later. As soon as the device is connected to the internet again, the App shall synchronize and replicate with the latest information, data and content.	

Category	Services	Classification (Mandatory / Optional)
	• Synchronization shall take place automatically without user intervention (unless specified differently)	
	• The App shall leverage push notification capabilities in order to remind users about upcoming events.	
Deliverables	• Data synchronization architecture • Definition of what is available online, offline and what data will be synchronized and when	
Acceptance Criteria / KPI (if applicable)	• Speed of synchronization for a defined data packet over different networks (e.g., 3G, LTE etc)	

Category	Services	Classification (Mandatory / Optional)
Topic	Interfaces to Enterprise systems and back-end services	O
Scope	• Describe how to integrate the App into the existing environment (processes, applications and infrastructure) including the corresponding technical and data integration concept and details of the required interfaces.	
	• Establish interface contracts (behavior definition and respective implementation agreement of callers and callees) with the stakeholders of all required Client backend and enterprise systems	
Deliverables	• Interface contracts with clearly defined inputs and outputs and their respective format definitions	
Acceptance Criteria / KPI (if applicable)	• Interface contracts approved by all stakeholders	

Category	Services	Classification (Mandatory / Optional)
Topic	Detailed Design Security and Data Privacy	M
Scope	• User authentication and authorization concept	M
	• Single-Sign-On specification	O
	• Transaction and session management (Browser, J2ME Clients) concept	M

Category	Services	Classification (Mandatory / Optional)
	• HTTP/HTTPS and data encryption specification	M
	• Transaction validity and application authenticity concept	O
Deliverables	• Detailed Security and Data Privacy Design • Threat model	
Acceptance Criteria / KPI (if applicable)	• Client to approve	

4.3 Implement – Develop

Category	Services	Classification (Mandatory / Optional)
Topic	Prototypes / mock-ups	O
Scope	• Generate detailed design and style guides for prototypes / mockups that adhere to the Client´s guidelines.	
	• Create and document a storyboard for the user interface interaction.	
	• Create clickable prototypes / functional mockups	
Deliverables	• Prototype / Mockup design • Prototype / Mockup	
Acceptance Criteria / KPI (if applicable)	• Prototype / mockup walkthrough	

Category	Services	Classification (Mandatory / Optional)
Topic	Development	M
Scope	• Develop mobile app source code according to defined standards as agreed with Client	M
	• Develop a back end solution using Client's mobile standard back end infrastructure (web, database, server)	O
	• Perform peer reviews and code walkthroughs as agreed by the Parties and report results of such peer reviews and code walkthroughs to Client.	O
	• Perform and document unit tests which cover all	M

Category	Services		Classification (Mandatory / Optional)
		Use Cases	
Deliverables	•	Mobile App deploy package, Source Code, Source Code documentation, peer review and code walkthrough reports	
Acceptance Criteria / KPI (if applicable)	• • •	Peer review and code walkthrough reports Test reports See Implement - Test	

4.4 Implement - Test

Category	Services	Classification (Mandatory / Optional)
Topic	Functional requirements and data flow	M
Scope	• Analyze each functional requirement of the prototypes and collect Client's feedback to finalize the functional requirements	O
	• Ensure and demonstrate that system requirements interacting models and wireframes (if applicable) are linked to business requirements	O
	• Validate the data flow of information and transition between screens	M
	• Validate application performance against requirements	M
	• Validate response messages during service outage	O
	• Check for the time taken to offline access the application services	O
Deliverables	• Performance report	
Acceptance Criteria / KPI (if applicable)	• RAM usage • Storage usage • Time needed in seconds	

Category	Services	Classification (Mandatory / Optional)
Topic	App specific testing	M
Scope	• Create a test plan and obtain Client approval	O
	• Create test cases for all non-functional requirements and obtain Client approval	O
	• Link system test cases to technical requirements	O
	• For each Use Cases create at least one test case	O
	• Conduct real device testing for all In-Scope-Devices	M

Category	Services	Classification (Mandatory / Optional)
	to ensure the mobile app provides consistent performance (e.g. CPU utilization, memory validation, etc.) for all test cases agreed with the Client	
	• Conduct stub based testing where interfacing applications miss or fail	M
	• Ensure and demonstrate tool based test management	O
	• Automate module testing for recurring build versions	O
Deliverables	• Test reports	
Acceptance Criteria / KPI (if applicable)	• Tested on In-Scope-Devices (or a suitable subset agreed by the Parties) • Allowed # of defects per device type; agreed implementation plan for all remaining defects	

Category	Services	Classification (Mandatory / Optional)
Topic	Security Testing	M
Scope	• Carry out Mobile specific testing including but not limited to areas like One Time User Password Verification, Session Management on Browser and J2ME Clients, HTTP/HTTPS over mobile browsers and Encryption Verification.	O
	• Consider security testing factors like Data Encryption, Transaction Validity, Application Authenticity, User Authenticity, and User Authorization.	M
	• The audience for the Apps may be B2B, B2C or B2E, and the security requirements needs to adhere to the industry standards to attract and retain this audience.	M
Deliverables	• Security Gap Analysis Report • Security Acceptance Report • Threat model • Security Test Plan and Test Cases • Assurance Document • Security Testing Report	
Acceptance Criteria / KPI (if applicable)	• Client to approve	

Category	Services	Classification (Mandatory / Optional)

Category	Services	Classification (Mandatory / Optional)
Topic	Test quality assurance	M
	• Ensure that the full code is covered by the unit tests	M
	• Document all conducted test cases	M
	• Check for system test cases links to technical requirements	M
	• Ensure and demonstrate tool based test management	O
Scope	• After the tests, conduct full test case review	O
Deliverables	• Test reports • Action items • Bug report	
Acceptance Criteria / KPI (if applicable)	• Conduct at least 3 cycles of testing to ensure comprehensive coverage • # of tests automated • RAM usage • Storage usage	

4.5 Operate - Publish

Category	Services	Classification (Mandatory / Optional)
Topic	Preparation for distribution	M
Scope	• Follow Client's standard procedure to prepare the mobile app for distribution	M
	• Export Xcode archive / application package	M
	• Compile required screenshots and icons	O
	• Notify Client of deployment package readiness (i.e. app binary (and screenshots where required))	M
	• Upon Client's provision of deployment method (e.g. upload link) provide deployment package	M
Deliverables	• Deploy package including documentation for publishing	
Acceptance Criteria / KPI (if applicable)	• Mobile app deployment package	

Category	Services	Classification (Mandatory / Optional)
Topic	App distribution	M
Scope	• Get approval from the Client to publish and – if agreed by Client - carry out the steps necessary to	M

Category	Services	Classification (Mandatory / Optional)
	publish the App to the relevant and selected App stores (e.g., Apple, Google and Windows Alternate:	
Deliverables	• Published App	
Acceptance Criteria / KPI (if applicable)	• Lead-time after the Client approval for App to be available on App store (delays due to Client internal approval steps shall be not considered)	

5.0 GENERAL SUPPLIER OBLIGATIONS

6.0 CLIENT PARTICIPATION

1. Client will provide:
    a. [optional]Project infrastructure on Client premises, e.g. workplaces and material for project work, subject to Client's discretion
    b. If owned by Client, access to documents, information and data, required for Supplier in order to perform the Services,
2. Additional obligations shall be agreed between the parties in written format prior to project start.

7.0 GENERAL SUPPLIER OBLIGATIONS

7.1 Cross-functional Obligations

Besides the requirements specified in Section 3.0 Supplier obligations shall include:

1. Creation and maintenance of a detailed project plan obtaining Client's respective approvals for the initial version and all updates.
2. Establishment of a requirements and change management process using an agreed tool.
3. Periodical status report of the project progress.
4. Obtain Client approval for each implementation phase as set out in section 3 of this document
5. Document all material provided for the Client on the basis of the project management methods agreed upon with Client. The documentation becomes the property of and is made available to Client (e.g. data models, source code, processes, APPLICATION user manuals, system administration manuals etc.)
6. Provide Quality Management and Quality assurance, subject to Client approval, designed to promote a high level of quality and focus on measuring and improving reliability, speed, cost-effectiveness, and Client satisfaction.
7. On request of Client, minutes for project management and governance meetings.
8. In particular organization and lead of all development-related meetings (e.g. scrums, sprint reviews, retrospectives, etc.).

9. On request of Client, plans to develop and adhere to the operating model in terms of sizing and engagement. For the proposed operating model, the Supplier should propose the locations, staff, roles and responsibilities for the Supplier core team and also list out expectations and touch points with the Client core team.

10. Comprehensive description of the proposed Application, Development Model including the Project Management Methodology, Engagement Model, Governance and any other aspect related to the Development of the App.

11. Demonstrate processes and standards for the execution of development, maintenance and support projects using Waterfall, Agile, Iterative, ITIL methodologies. During Initiation phase, based on the chosen development methodology, related processes and standards gets included in the project life cycle and is tracked throughout the life cycle of the project.

12. The code of the App should be well documented applying the standard documentation practices for App development or any Client standards that may be applicable.

13. The copyright for the App and contents (text, images, multimedia) that have been created for the Client must be assigned to the Client.

14. The Client may be required to purchase any applicable third party licenses for any third party products that are necessary for the Supplier to design and develop the Product. Such third party products may include, but are not limited to: server-side applications, clip art, "back-end" applications, music, stock images, or any other copyrighted work which the Supplier deems necessary to purchase on behalf of the Client to design and develop the Product. In the event any such third party product exceeds €xxx.xx per product (or €xxxx.xx in aggregate), the Supplier shall obtain the Client´s prior written consent before incorporating such third party product into the Product. The Supplier shall provide the Client with a list of all third party products before deployment of the App.

7.2 People, Tools and Standards

3. The Supplier shall ensure that people working on the application have a sound know-how of regulatory systems compliance and standards (e.g., ISO 13485, ISO 14971, 510(k)/PMA/DMF submission, CSV, GxP, 21 CFR Part 11) as well as quality and documentations standards.

4. The Supplier shall ensure that quality systems and development processes are also compliant to medical standards (e.g., FDA, HIPAA if applicable)

7.3 Source Code & IPR

1. Source code is the property of the Client.

2. The Supplier will assign to the Client, without further compensation, all of its right, title and interest in and to the App and any and all related patents, patent applications, copyrights, copyright applications, trademarks and trade names.

3. The Supplier will keep and maintain adequate and current written records with respect to the App (in the form of notes, sketches, drawings and as may otherwise be specified by the Client), which records shall be available to and remain the sole property of the Client at all times.

4. All versions of the App shall contain the Client´s conspicuous notice of copyright.

5. The Supplier will assist the Client in obtaining and enforcing patent, copyright and other forms of legal protection for the App in any country.

6. Upon request, the Supplier will sign all applications, assignments, instruments and papers and perform all acts necessary or desired by the Client to assign the App fully and completely to the Client and to enable the Client, its successors, assigns and nominees, to secure and enjoy the full and exclusive benefits and advantages of this work.

7. The term “intellectual Property Rights” means all copyrights, patents, trademarks, trade secrets, proprietary rights and other intellectual property rights, including the right to grant a license in the form hereof. The Supplier warrants that the App will not infringe upon any copyright, patent, trade secret or other intellectual property interest of any third party. The Supplier will indemnify and hold the Client harmless from and against all such infringement claims, losses, suits and damages including, but not limited to, attorney's fees and costs, and shall promptly following any bona-fide claim of infringement correct the App so as not to be infringing, or secure at its own expense the right of the Client to use the App without infringement.

8. The Client shall have transferable, exclusive licence and rights to use, distribute and make available either free or with charges the App to anyone including employees, subsidaries, affiliates, patients, medical practictioners and customers. No licence fee or any kind of additional remuneration will be charged by the Supplier for this.

Globalization[optional]

1. The Client´s Apps will be deployed and run in different markets. Therefore, localization strategy will be needed. The Supplier should propose an efficient localization strategy.

7.4 Security and Compliance

1. The Supplier’s responsibilities will include:
    a. Implement all security requests associated with App code and executable modules subject to Bayer approval on all data or information requests.
    b. Install and maintain source control Software.
    c. Monitor and restrict access to source code and data.
    d. Comply with Ad Hoc, annual audits, and regulatory requests.
    e. Report any security violations.
    f. Adhere to industry specific IT security and Data protection standards

7.5 Compliance

1. Compliance to regulatory (FDA and non-FDA) requirements shall be provided by the Supplier and the Supplier´s approach should at least cover the following:
    a. Review the List of regulatory requirements in regulatory compliance control
    b. Break the controls into Process level controls, infrastructure level controls and application level controls
    c. Update the process documents to ensure that the controls are implemented
    d. Update the SDLC to ensure that the application level controls and Infrastructure level controls are documented in the requirements and design guidelines.

e. Ensure that the traceability matrix takes care that the application level requirements are documented, designed for, tested and validated though out the SDLC

f. The set of deliverables depends on the regulation under consideration.

7.6 Milestones and Project Plan

1. Supplier shall promptly notify Client in case of any delay of project milestones or deliverables. For each delay Supplier shall provide written notice to Client on the reasons of the delay and expected delay along with proposals for mitigation. For the avoidance of doubt, this does not release Supplier from its duties and obligations under this project agreement. Each prolongation of deadlines under this agreement requires Client's written approval.

2. In case of Supplier failure to meet his duties and obligations under this project agreement, Client shall have the rights set forth in the law.

7.7 Locations

1. The following regulations regarding project locations shall apply:

   a. Project meetings shall take place on Client premises provided that Client has not agreed to Supplier to attend via Phone, Video or Web Conference.

   [Alternative] Services shall be solely provided on Supplier premises

   [Alternative] Services shall be solely provided on Client premises

   [Alternative] Service shall predominantly be provided on Supplier premises. Project activities listed below shall be delivered from Client premises:

      i. Kick-off

      ii. User Acceptance Test

2. Supplier shall not invoice Client for travel cost to agreed Client project locations. Cost related to travel to other Client locations requires Client pre-approval. Client travel expense policy in effect applies.

8.0 SERVICE LEVEL

1. In addition to the KPIs and Acceptance criteria described in Section 3.0 the following Service levels shall apply for the purpose of this SOW.

8.1 On-Time Delivery of Project Milestones

1. The Service Level „On-Time Delivery of Project Milestones“ measures the level of project milestones completed on time. As a minimum, the end of project phases defined in Section 3.0 Requirements and App Scope are agreed as milestones.

2. The metric for this Service Level is calculated as follows:

   Number of all project milestones since project start delivered on or before the scheduled date divided by the number of project milestones scheduled to be completed since project start * 100 [%]

3. In case Client has requested or approved a postponement of milestones, the approved new milestone dates applies for the Service Level measurement.
4. With each milestone Supplier shall report the achieved Service Level to the Client.

8.2 On-Time Delivery of Project Reports

1. The Service Level „In Time Delivery of Project Reports“ measures the level of agreed project reports delivered on time.
2. The metric for this Service Level is calculated as follows:

   Number of agreed project reports delivered on time since project start divided by the number of agreed project reports due since project start * 100 [%]

3. With each milestone Supplier shall report the achieved Service Level to the Client.

8.3 Number of Milestones within the first User Acceptance Test (UAT)

1. The Service Level „Number of Milestones within the first User Acceptance Test (UAT)“ measures the level of agreed milestones which had been accepted/approved during the first UAT (while first Acceptance procedure).
2. The metric for this Service Level is calculated as follows:

   Number of milestones delivered with a 100% faultless first User Acceptance Test (UAT) divided by the total number of milestones provided by supplier for client UAT since project start * 100 [%].

3. With each milestone Supplier shall report the achieved Service Level to the Client.

8.4 Client Satisfaction Survey

[optional]

8.5 Service Level Agreements (SLA)

1. The following Service Level and KPIs are defined:

Service Level Matrix				
Ref. #	Service Level Name	Expected KPI	Minimum KPI	Measurement period
1	On-Time Delivery of Project Milestones	100%	80%	Monthly
2	On-Time Delivery of Project Reports	90%	80%	Monthly
3	Number of Milestones within the first User Acceptance Test (UAT)	100%	90%	Monthly
4	Client Satisfaction Survey			

[optional]

2. If any of the above defined and agreed Service Level is below the Minimum KPI for any one (1) measurement period or if it is below Expected KPI for three (3) consecutive measurement periods, the Supplier shall provide Client with a written plan for improving the respective Service Level within fifteen (15) calendar days of failure to meet that Service Levels.

[optional – please adjust]

3. The following rules for failure to achieve service levels will apply:

   a. If the Supplier fails to maintain the Service Level Times, a reduction of the monthly fee of EUR XXX will apply. The reduction per calendar month shall be limited to EUR XXX.

   b. This rule shall apply after the XX per month XX.

   c. In case that the error cannot be remedied through a bug fix or a hotfix, but instead the complete redevelopment of a function or part of a module is required, then this rule shall not apply thereto.

9.0 COMPENSATION

9.1 Fixed price

1. Client shall pay a fixed price of ….. EUR for all Services according to Section 3.0 delivered by Supplier.

2. The Payment of the fixed price is due:

   a. upon final acceptance and completion of the project

   b. according to below specified payment schedule

Payment Due Date	Date of Completion	Amount
Completion milestone 1	DD.MM.YYYY	xx.xxx,xx €
Completion milestone 2	DD.MM.YYYY	xx.xxx,xx €
Completion milestone 3	DD.MM.YYYY	xx.xxx,xx €
Completion milestone 4	DD.MM.YYYY	xx.xxx,xx €
Completion milestone 5	DD.MM.YYYY	xx.xxx,xx €
Total		**xxx.xxx,xx €**

   c. … (insert alternative options)

9.2 Other cost

1. No other cost will be incurred.

10.0 TERM AND TERMINATION

1. This project agreement shall take effect as of ...., but earliest as of approval of both Contract Partners (Effective Date).

2. The term of the project agreement will be …, starting with the Effective Date.

3. Client may terminate the SOW, in whole or in part at month end within one (1) month written notice. In case of termination Client shall notify Supplier which Services shall be performed by Supplier. Supplier shall deliver these Services under the terms and conditions of this SOW. Supplier will in this case be entitled to payment for all agreed Services delivered until termination.

4. Provisions of the MSA for termination shall apply.

11.0 SIGNATURE

Authority to proceed

Leverkusen, ...	XYZ, ...
....	....
(Client)	(Supplier)

ANNEXURE 1 ZUM RAHMENVERTRAG: VORLAGE

Verpflichtungserklärung

bezüglich der Wahrung des Datengeheimnisses, Bankgeheimnisses, der Betriebs- und Geschäfts-geheimnisse sowie der Nutzung der elektronischen Bürokommunikations- und Informationsmittel des Auftraggebers.

1 Ich, ____, verpflichte mich

(Vorname, Name des Mitarbeiters des Auftragnehmers)

1.1 (*Datengeheimnis gem. § 5 Bundesdatenschutzgesetz/BDSG*) bezüglich aller durch das BDSG geschützter personenbezogener Daten, die bei dem Auftraggeber verarbeitet werden und von denen ich im Rahmen meiner Tätigkeit für den Auftragnehmer Kenntnis erlange, stets das Datengeheimnis zu wahren. Aufgrund des Datengeheimnisses ist es mir untersagt, diese Daten unbefugt zu einem anderen als dem zur jeweiligen rechtmäßigen Aufgabenerfüllung gehörenden Zweck zu verarbeiten, Dritten bekannt zu geben oder zugänglich zu machen oder sonst zu nutzen. Ich werde diese Daten daher ausschließlich entsprechend den Weisungen des Auftragnehmers verwenden.

1.2 (*Bankgeheimnis*)
bezüglich aller mir im Rahmen meiner Tätigkeit für den Auftragnehmer bekannt gewordenen Rechtsbeziehungen des Auftraggebers zu ihren Kunden stets das Bankgeheimnis zu beachten. Ich werde daher Dritten gegenüber hierüber absolutes Stillschweigen bewahren.

1.3 (*Betriebs- und Geschäftsgeheimnisse des Auftraggebers*)
bezüglich aller sonstigen Informationen, die entweder ausdrücklich oder der Natur der Sache nach zum vertraulichen internen Gebrauch bestimmt sind, ebenfalls stets absolute Vertraulichkeit zu wahren.

1.4 (*Bürokommunikations- und Informationsmittel*)

die vom Auftraggeber vorgegebenen Büro- und Informationsmittel ausschließlich nach den Bestimmungen *über die Nutzung von elektronischen Bürokommunikations- und Informationsmitteln in der Bayer AG* zu nutzen.

2 Die Verpflichtungen gem. vorstehender Ziffern 1.1 - 1.4 werde ich auch nach Beendigung meiner Tätigkeit für den Auftragnehmer beachten.

....

(Ort/Datum) (Unterschrift des Mitarbeiters)

ANNEXURE 2 BAYER WEBGUIDE

<Note to the BBS Service Owner: Insert latest webguide here>

ANNEXURE 3 GLOSSARY

Term	Characteristic / Definition
Android	Linux-based operating system designed primarily for touch screen mobile devices such as smartphones and tablet computers. Originally developed by Android, it was purchased by Google in 2005. Android is open source and used by numerous device manufacturers on their devices.
Android application package file (APK)	A file format that is used deliver mobile apps to Android devices
App builds	A process of compiling your mobile app code into a finished mobile app product. it's a packaged/compiled version of your mobile App.
App developer	An individual that creates apps for mobile devices
App developer team	A group of people involved in the mobile app development process. can include project managers, product managers, marketers, game producers, mobile app designers, and mobile app engineers.
App distribution	Process of getting your mobile app to end users typically through a mobile app store such as the Google Play Market and the Apple App Store
App icon	Represents your mobile application in the App store
App Store	Termed "App Store" for Apple Device applications, "Market" for Android and "Blackberry App World" for Blackberry apps. An App Store is an online store from which mobile device users can obtain applications. Developers submit mobile applications to app stores for distribution to end users.
App Store Optimization (ASO)	Process that helps increase your visibility in the app stores with a focus on optimizing your app content
App store rank	Your position in the app store based on downloads, sales and usage
App store rating	Numeric measurement mobile app users assign to your mobile application as a quality indicator of your mobile application. These app store ratings are submitted to mobile app stores for future users to use as a screening measurement before downloading your mobile app.
App user	A person that downloads and uses your mobile app.
Application Distribution	The process of submitting an app to an app store for review and distribution to public consumers or internally within enterprises. Apple has strict guidelines for app distribution; one must have either a developer's license to distribute apps to public consumers or an enterprise license to distribute apps internally within enterprises. Android, Blackberry, and Windows phone platforms have a more lenient review process, although they also have licensing requirements.

Application Provisioning	Refers to the process of creating a profile provision on the iOS platform for a specific app. The provision contains two major elements, the app ID and a list of devices (only if an ad hoc provision or a development provision, otherwise a list is not required for app store submission).
Authentication	Refers to the verification of a user's identity with something the Mobile App User provides, such as date of birth, password, secret question, a smart card (physical item), fingerprint (physical feature).This is also a method for verifying sensitive, private, and/or premium data.
B2B	Business to Business
B2C	Business to Consumer
B2E	Business to Employee
Badge	Refers to the iOS platform, a badge is the visual notification bubble that alerts the Mobile App User to the number of unread emails.
Beta testing	Process of testing a pre-release version of your mobile app
Bluetooth	Is a wireless technology standard for exchanging data over short distances from fixed and mobile devices.
Bugs	An unintended behavior in your mobile application
Certificate Signing Request (CSR)	In preparing a device for development on the iOS platform, a CSR is a needed digital asset containing personal information, which is used to generate developer certificates.
Churn rate	How frequently mobile app users abandon your app after a certain period
Content (in mobile app terms)	Anything that is not a design or programming element in your mobile app is content. any text, images, audio, video, HTML, etc. that the mobile app users can see in your app would be considered content.
Debugging	A process of identifying, tracing, and resolving bugs in a mobile application
Developer Certificate	An additional needed digital asset, which identifies an IOS application developer; after the CSR approval, the developer certificate is downloaded from the iOS portal, and then added to the developer's keychain.
Device Orientation	Directional presentation of mobile device contents based on the physical orientation of the device - landscape (wide) and portrait (tall).Apple, Android, and Windows Phone 7 devices are capable of rendering screens in landscape and portrait views. Differing views of application contents are often presented based on device orientation.
Direct Manipulation	Refers to or describes interactive systems where users physically interact with their operating system offering "user control." Instead of typing text commands, the Mobile App User has the ability to physically interact with directories and files while receiving visual representation of the progress and the end point. An example of direct manipulation is

	the resizing of a graphical shape such as a rectangle box.
Download velocity	How often a mobile app is downloaded in a certain time frame
Enterprise mobile apps	A mobile app that provides functionality primarily to enterprise business and organizations. Typically the app solves user control, role based access, and caters toward the scale of enterprise workplaces.
Expedited app review	A review process performed by Apple that gets your iOS app approved for the app store quicker. this review process is reserved for updates with major bugs or releases that are time sensitive. Apple only allows a couple per account and in rare cases.
Gestures - Pinch w/ 2 fingers	A direct manipulation action where a user places two fingers on a mobile device screen and moves them toward each other. The displayed content is made smaller in response.
Gestures - Pinch w/ 4 or 5 fingers	A direct manipulation action where a user places for or five fingers on a mobile device screen and moves them toward each other. The displayed content is replaced by the home screen.
Gestures - Rotate	A direct manipulation action in which the Mobile App User moves two or more fingers in a circular motion on a mobile device screen. In response, the displayed content will rotate/change orientation.
Gestures - Shake	A direct manipulation action used on iOS mobile devices in which the Mobile App User shakes the device. The device will perform a variety of actions in response, depending on the application in use.
Gestures - Spread w/ 2 fingers	A direct manipulation action where a user places two fingers on a mobile device screen and moves them away from each other. The displayed content is made larger in response.
Gestures - Swipe w/ 1 Finger	A direct manipulation action, also known as a ‘flick’, where a user wipes a finger across the mobile device screen, pushing the displayed content in the direction of the wipe.
Gestures - Swipe w/ 4 or 5 Fingers Horizontal	A direct manipulation action used on an iPad in which the Mobile App User places four or five fingers on the screen and wipes horizontally. In response, the iPad will scroll through the most recently used applications one by one.
Gestures - Swipe w/ 4 or 5 Fingers Vertical	A direct manipulation action used on an iPad in which the Mobile App User places four or five fingers on the screen and wipes upward. In response, the iPad will present the multi-tasking bar at the bottom of the display
Gestures - Tap / Double	Direct manipulation action in which the Mobile App User taps the mobile device screen twice. In response, the displayed content zooms larger or smaller, depending on the zoom state when the double tap is executed. If the content is not zoomed,

	it will become larger. If it is already zoomed, it will become smaller.
Gestures - Tap / Single / aka Short Touch	A direct manipulation action where a user touches the mobile device screen quickly. If there is an actionable element present where the Mobile App User touched, the action will be executed.
Gestures - Tap and hold / aka Long Touch	A direct manipulation action where a user touches the mobile device screen for several seconds. If there is an editable element present where the Mobile App User touched, the touched content will be highlighted and editing options will be presented onscreen for user selection.
Gestures - Tap, hold, and drag	A direct manipulation action where a user touches the mobile device screen for several seconds. If there is an editable element present where the Mobile App User touched, the touched content will be associated and highlighted. The Mobile App User may then drag the cursor to the position in which they wish to begin editing. On release of their finger, they will be able to edit the highlighted content.
HTML5	Hyper-Text Mark-up Language Version 5 - An internet browser language used for development of mobile applications. HTML5 based applications can be utilized on any mobile device with internet browser capability.
I18n	An abbreviated numeronym of the word internationalization where 18 stands for the number of letters between the first i and last n of the word
In-app ads	Advertisements that display within your mobile app to your mobile app users
In-app messages	Any message that originates from inside the app through means other than push notifications
In-app Purchase (IAP)	Purchases of items and functionality within your mobile pap
In-House App Store	A proprietary store used to distribute an organization’s mobile apps to its users. Developers submit mobile applications to the app stores, where employees can browse, download, and install various apps.
Internal mobile apps	Mobile apps that are typically built for internal use at companies. Internal apps
Internationalization	(See Localization)
iOS	The Operating system that runs on the iPhone or iPad.
Iterations	A different or updated version of your mobile app
Locales	A group of particular users with similar geographic, linguistic, and grammar patterns. Usually used to organize mobile app users in groups for app localization.
Localization	Customization of content for different language speakers. Localization can include translating your mobile app content text, images, video, and more for a particular local market.

Location Based Services (LBS)	A service that utilizes a user's geolocation to provide functionality.
minnows	Users that don't spend much in your mobile app. primarily used to refer to small spenders in freemium apps
Mobile App	A mobile application or "mobile app" is defined as a software application that can be executed (run) on a mobile platform (i.e., a handheld commercial off-theshelf computing platform, with or without wireless connectivity), or a web-based software application that is tailored to a mobile platform but is executed on a server
mobile app a/b testing	Process of testing two or more variations of app element like buttons, background colors, fonts, screens, and app content.
mobile app analytics	Way to track the actions of mobile app users within your app
mobile app API	The interface or set of functions that are available to through developer through the framework or SDK. It's basically the documentation of things you can do with an SDK. It helps you understand the rules or guidelines that tell you what you can and can't do within a framework.
Mobile app content delivery network (CDN)	An application that helps you deliver your mobile app content.
Mobile app content management system (CMS)	An application that helps you publish, deliver, and update mobile app content sent to mobile app users from one central place.
mobile app feature switching	Aka feature toggle. Feature Flag, Feature Flipper, Conditional Feature, etc the capability to turn on and off specific functionality in your mobile app remotely, without coding and breaking other parts of your mobile app.
mobile app sdk	Self-contained library or framework which allows developers to extend the functionality of the library into their mobile application. For instance, the Facebook SDK allows you to add Facebook features right into your app.
mobile app user retention	How many users have your mobile app and consistently use it.
Mobile Medical App	A "mobile medical app" is a mobile app that meets the definition of device in section 201(h) of the Federal Food, Drug, and Cosmetic Act (FD&C Act) - 7 - 4 ; and either is intended: 1) to be used as an accessory to a regulated medical device; or 2) to transform a mobile platform into a regulated medical device.
Native app	Mobile app built to work specifically for one platform
Native Application	An application created in the development language created for specific to a particular mobile device.
Over-the-air (OTA)	A download that initiates over a wireless broadband connection from a mobile device

Push fatigue (aka push notification fatigue)	A decline in mobile app user interest in push notifications from the overuse and that has lead to lower user engagement gains
Push notification	A short message mobile app developers can send to app users even when users don't have their mobile applications open.
Retina icon	An app icon that is saved at double it's size for optimal viewing on retina-enabled mobile devices. creating a retina version of your mobile icon will deliver a crisp version of your mobile icon to users that view your icon on retina devices
Selection of Dev Approach in Mobile Apps	Agile works well if the business is willing to be involved closely in the project during the entire project phase. Waterfall approach works well when business has limited time or if it is involving multiple countries and stakeholders.
SMS	A format used for sending text between mobile devices
Thick Client	Applications created in an operating system created in a language specific to a particular mobile device. At PwC, iOS devices are the standard mobile device. Applications created in the iOS environment at PwC are referred to as 'Thick Client' applications.
Thin Client	Browser based applications developed for use on mobile devices. Thin client applications are developed using html5 and are typically unable to deliver the rich user experience available when developing in the operating system native to the a particular mobile device.
Unique Device Identifier (UDID)	A unique alphanumeric that identifies a specific mobile device. Typically is a number assigned by the mobile device manufacturer.
Universal app	A mobile app that can run on multiple devices using the same platform. for example, a mobile app that can run on any device (the iPod, iPad, and iPhone) on the iOS platform
User engagement	How frequently your mobile apps open and interact with your mobile app. Important metric that correlates with user retention
User Experience	The way a person feels about using a product, system or service. User experience highlights the experiential, affective, meaningful and valuable aspects of human-computer interaction and product ownership, but it also includes a person's perceptions of the practical aspects such as utility, ease of use and efficiency of the system. User experience is subjective in nature, because it is about an individual's feelings and thoughts about the system. User experience is dynamic, because it changes over time as the circumstances change.

User Interface	The mechanism by which people (users) interact with a machine. The Mobile App User interface includes hardware and software components. User interfaces exist for various systems, and provide a means of: 1) Input, allowing The Mobile App User to manipulate a system (ex, keyboard, touch screen, buttons) 2) Output, allowing the system to indicate the effects of The Mobile App User' manipulation (ex, computer monitor, mobile device screen)
Whales	The bigger spenders in mobile apps. These users tend to spend a lot on in-app purchases in mobile apps. They also generate the most revenue for app developers.
Wireframes	A design draft or prototype of the functionality and navigation of a mobile app. Typically created when developing a new mobile app or add a feature to an existing app.

ANNEXURE 4 USE CASE TEMPLATE

<Note to the BBS Service Owner: Each mobile app function should be described by a separate Use Case with a self-explanatory name. The Use Cases can be either attached to this document as a separate document or as Annexure 5 within this document in both cases with a section for each Use Case.>

Use Case name	
Display	ID to define display in display tree structure
Goal	Brief description (or reference to respective User Story)
User group(s)	Only applicable if Use Case refers to a sub group of user groups described in part 1 (General Description)
Business processes supported	•
Preconditions	Define all the conditions that must be true (i.e., describes the state of the system) for the *trigger* (see below) to meaningfully cause the initiation of the use case. That is, if the system is not in the state described in the preconditions, the behavior of the use case is indeterminate. Note that the preconditions are *not* the same thing as the "trigger" (see below): the mere fact that the preconditions are met does NOT initiate the use case. In seldom cases, a condition is both precondition and trigger. Preconditions should not list the possibly preceding use case – this would be a trigger.
Actor	Person, device or system that • sets off the trigger (see below) • acts on the system • interacts with the system during the course of events (see below)
Trigger	• Describe the external or internal event that causes the use case to be initiated. • Can be a single event (actor presses a button, data element has changed) or a set of conditions to be met (if the last is true, prove whether preconditions and trigger are the same) • If several conditions should set off the use case, a monitoring trigger process should be designed/defined together with is a signal from the trigger process when the conditions are met. If the trigger occurs but the preconditions are not met: • Strictly this should not be possible, if the preconditions are not met, the trigger cannot set off the use case (e.g. user cannot press button) • Can be handled as an error/exception (alternative paths) (see below)) of the use case • The preconditions can define the trigger (so that the use case does not

	run if the preconditions are not met) - create a different use case to handle the problem. (If this is the common case, a preconditions section is unnecessary.
Course of events	Primary scenario or typical course of events often described as a set of usually numbered steps. For example: 1. System prompts the user to log on, 2. User enters his name and password, 3. System verifies the logon information, 4. System provides welcome screen.
Alternative paths	• Secondary path or alternative scenarios, which are variations on the main theme. Alternatively use conditional logic or activity diagrams to describe use case with many rules and conditions. • Exceptions, or what happens when things go wrong at the system level, may also be described (see above), not using the alternative paths section but in a section of their own. • Alternative paths are numbered in relation to the basic course of events to show at which point they differ from the basic scenario, and, if appropriate, where they rejoin. The intention is to avoid repeating information unnecessarily. The description of an exception should indicate how the system will respond to, or (if possible) recover from, the error condition.
Post conditions	Describes what the change in state of the system will be after the use case is completed. Post-conditions are guaranteed to be true when the use case ends.
Dependence on other Use Cases	Could also be handled by pre- and post conditions
Comments, references	
Pending items	

ANNEXURE 5 USE CASES

<Note to the BBS Service Owner: In case the Use Cases are attached as a separate document this Annexure shall only refere to that document name.>